

Report — Proportional Control for Maneuvering Flapping Flight

Jean Pecquet

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1 Model Description

We model a dragonfly-like insect with two pairs of wings. The sagittal plane is aligned with the fixed frame (x, z) plane, where z points opposite the direction of gravity. The wings in each pair beat symmetrically about the sagittal plane, and there is zero initial velocity or wind component normal to the sagittal plane, so the motion is constrained to the (x, z) plane, rendering the problem two-dimensional. A subsequent strong simplification is that the body is reduced to a point mass, so there is no coupling between body pitch and wing kinematics. In this manner we can focus entirely on the translational dynamics without concern for flight stability. The dynamical model is nondimensionalized using the body length L , the body mass m , and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the wing kinematics angular parameters, are listed in Table 1. Dimensionless quantities are denoted by an asterisk (*).

Each wing is modeled as a single, massless plate element with its aerodynamic center at the $\frac{2}{3}$ -span station. The lift and drag coefficients of each element are prescribed as a function of the instantaneous angle of attack α (provided at the end of the section). The wing kinematic angles are represented in Figure 1. The wing flapping angle $\phi(t)$ and pitching angle $\psi(t)$ are prescribed by sinusoidal functions as follows

$$\begin{aligned}\phi(t^*) &= \phi_1 \sin(\omega^* t^*) \\ \psi(t^*) &= \psi_0 + \psi_1 \sin(\omega^* t^* + \delta_0)\end{aligned}$$

The forewing angles are exactly the above, while the hindwing angles have an additional phase parameter σ_0 added to the sin phase. Table 1 lists the nondimensionalized parameters in the dynamic model. Ranges for the parameters L_w^* , A_w^*/m^* , and ω^* were obtained from the morphological data in [1] and the wingbeat frequency data in [2] for *Anisoptera*.

Table 1: Dimensionless parameters

Parameter	Description	Range
$L_w^* = \frac{L_w}{L}$	Wing length ratio	[0.65, 0.85]
$A_w^*/m^* = \frac{A_w \rho_{\text{air}} L}{m}$	Inverse wing loading (per wing)	[0.05, 0.25]
$\omega^* = \omega \sqrt{\frac{L}{g}}$	Wing beat frequency	[8, 20]
γ_0	Wing stroke plane angle	[-180°, 180°]
ϕ_1	Wing flapping amplitude	[0°, 60°]
ψ_0	Mean wing pitch angle	[-60°, 60°]
ψ_1	Wing pitch amplitude	[0°, 60°]
δ_0	Wing pitch phase	[-180°, 180°]
σ_0	Fore/hind phase	[-180°, 180°]

Lift and drag coefficients are given by

$$C_L = C_{L,0} \sin 2\alpha$$

$$C_D = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha$$

with $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, $C_{D,\frac{\pi}{2}} = 2$.

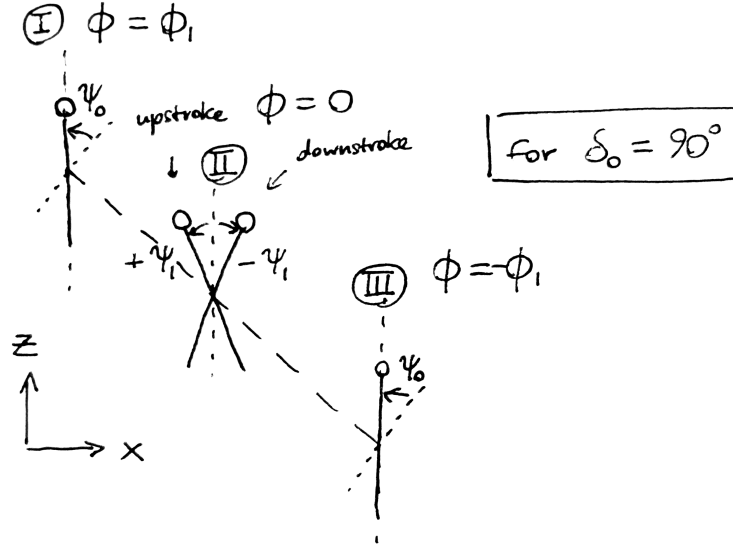


Figure 1: Wing kinematic angles for $\delta_0 = 0$, represented on one wing for clarity [placeholder]

2 Stationary Flight

Main results:

- Effect of pitch parameters $(\psi_0, \psi_1, \delta_0)$ on mean aerodynamic force is separable from morphological parameters and wingbeat frequency
- Pitch parameters $\psi_0 = 0^\circ$, $\psi_1 \approx 51^\circ$, and $\delta_0 = 90^\circ$ maximize mean aerodynamic force given $(\frac{A_w^*}{m^*}, L_w^*, \omega^*)$
- A two-variable proportional scheme using ϕ_1 for force magnitude and ψ_0 for force direction is sufficient to achieve stable hovering
- Inclined stroke plane hover exhibits increased damping in the vertical direction compared to a horizontal stroke plane

2.1 Equilibrium Solutions

A good first step for mapping the parameter space might be to identify those parameter values that give an average aerodynamic force over a cycle that is equal to or greater than the weight, i.e. $F_a^* = \frac{F_a}{m_b g} \geq 1$. Outside of this group of solutions, hovering and meaningful maneuvering flight cannot be achieved. This is independent of the stroke plane angle γ_0 and the fore/hind phase shift σ_0 , which limits us to just 7 parameters.

We can further eliminate one parameter by collapsing L_w^* and ϕ_1 into a single wing displacement amplitude parameter $s_0^* = \frac{2}{3} L_w^* \phi_1$ (displacement at $\frac{2}{3}$ -span station, aerodynamic center) resulting in 6 morphological and wingbeat parameters.

Verification Against Analytical Solution

As a first, quick verification we compare the $F_a^* = 1$ contour result in the $(A^*/m^*, \omega^*)$ plane to the simplified analytical result of [3] for a translating element with constant 45° pitch through each beat half:

$$(F_a^*)_{\text{constant pitch}} = \frac{C_{L,0}}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \quad (1)$$

Note $A^* = 4A_w^*$. We compare our model with the above expression at the reference condition from [3]: $s_0^* = 1.5$ (achieved in practice with a small ϕ_1 and a large L_w^*), $\psi_0 = 0^\circ$, $\psi_1 = 45^\circ$, and $\delta_0 = 90^\circ$. Crucially we do this assuming zero body velocity: the velocity of the wings relative to the air is equal to the velocity of the wings relative to the body. Contours are shown in Figure 2. Using a single wing element at the $\frac{2}{3}$ -span station, there is very close agreement, as expected, with a small difference attributable to the sinusoidal wing pitch (as opposed to a square wave). The wing element spends less time at large angle of attack so the mean force is slightly lower. As the number of elements is increased F_a^* converges to a lower value (by about 25%), causing the contour to shift towards higher A^*/m^* and ω^* . We use a single element at $\frac{2}{3}$ -span throughout for simplicity.

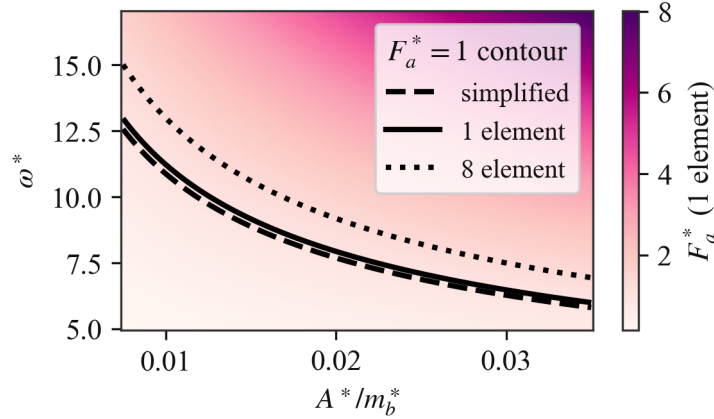


Figure 2: Verification of $F_a^* = 1$ contour at reference condition

Separating the Effects of the Parameters on F_a^*

The present model allows for more complex wing motions than a simple symmetric stroke: the mean pitching angle ψ_0 may be nonzero, and the phase shift δ_0 between the pitching and flapping motions may be different from 90° . One question we can ask is whether the contribution of the pitch to the force magnitude F_a^* can be represented as a coefficient that would depend on $(\psi_0, \psi_1, \delta_0)$, independent of the other 3 parameters. Looking back to the simple analytical expression for F_a^* for constant pitch, we can extract the dimensionless dynamic pressure

$$q^* = \frac{1}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 = \frac{A_w^*}{m^*} (s_0^* \omega^*)^2 \quad (2)$$

We can then evaluate F_a^*/q^* and check whether it is independent of $(\frac{A_w^*}{m^*}, s_0^*, \omega^*)$ over the parameter space, i.e. whether the two groups of variables can be cleanly separated. Again, we do this first at zero body velocity, such that $F_a^* = 1$, applied in the right direction, would result in a hover. As can be seen in Figure 3, the separation is clean, with small amounts of scatter associated with the nonlinearity of the flapping motion. The right side of the plot is obtained by picking a reference amplitude ($s_0^* = 0.25$, or $\phi_1 \approx 30^\circ$) to evaluate F_a^* . The sensitivity to the choice of s_0^* is very small, about 1 to 2%. The result is completely independent of A_w^*/m^* and ω^* because they appear purely as multiplicative factors in the analytical expression. It might be clearer to do a purely analytical treatment of this part.

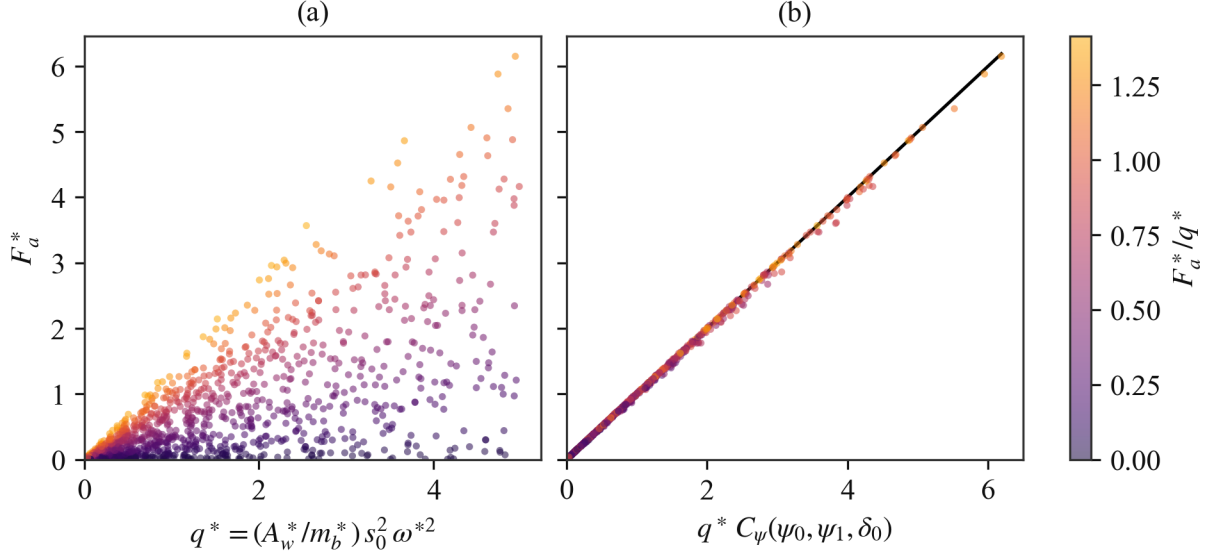


Figure 3: Separation of dynamic pressure and wing pitch factors of the aerodynamic force at zero body velocity. (a) Scatter of mean aerodynamic force as a function of dynamic pressure for random sampling of parameter space (b) Collapse of scatter by introduction of C_ψ parameter evaluated as F_a^*/q^* for a reference amplitude value $s_0^* = 0.25$

Optimal Wing Pitch Parameters

This decomposition appears to be useful because C_ψ provides a measure of the power efficiency of the wingbeat. A given net force can be achieved with a faster or slower wingbeat depending on the wing pitch pattern. If the pattern is such that C_ψ is high, then the wing doesn't have to beat as fast, and less energy is expended (provided any additional energy required to impart that particular pitch motion does not cancel out this benefit, which seems like more of an edge case). Presumably then, values of ψ_0 , ψ_1 , and δ_0 that result in higher C_ψ would be preferred in nature. Looking again at the simplified case from earlier it seems the optimum should sit near $\psi_0 = 0$, $\psi_1 = 45^\circ$, $\delta_0 = 90^\circ$, since a 45° angle of attack maximizes lift (limit case for square wave pitch: $C_\psi = C_{L,0}$).

We show the dependence of C_ψ on the three parameters in Figure 4. Again, C_ψ here is obtained by taking $s_0^* = 0.25$, but this value has little effect on C_ψ as seen in the scatter plot from earlier. The result is what we expect, with ψ_1 slightly larger than 45° to provide a mean angle of attack that is closer to 45° . The dependence of F_a^* on δ_0 is symmetric but the force direction depends on the sign, we highlight $\delta_0 = +90^\circ$ since with our conventions it results in a force in the direction of the leading edge. This optimal value of C_ψ is slightly smaller than $C_{L,0}$. Note that because the optimal pitch pattern is symmetric, the drag coefficient

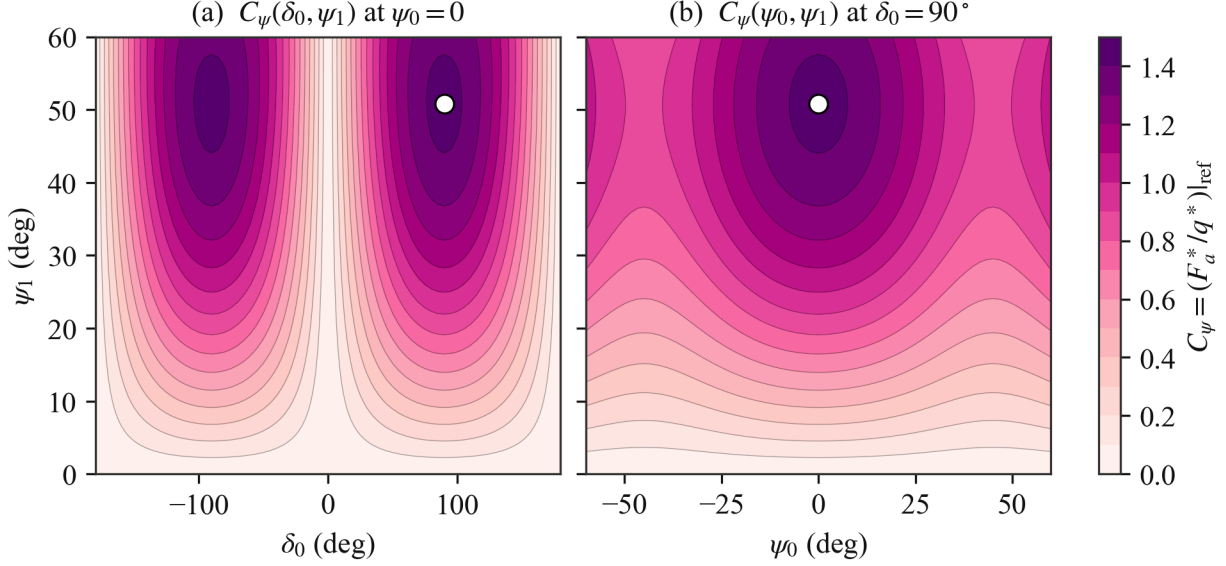


Figure 4: Dependence of C_ψ on pitch parameters at zero body velocity. Optimal value shown as white dots.

has no effect (drag component cancels out over the two halves of the wingbeat). We notice that with more wing elements, the optimal C_ψ is lower, converging to a value just above unity. This suggests the following approximate criterion for hover feasibility, which may also extend to maneuverability

$$\frac{A_w^*}{m^*} (s_0^* \omega^*)^2 \gtrsim 1 \quad (3)$$

The corresponding 3D surface is shown in [Figure 5](#). The required wingbeat amplitude ϕ_1 for a given $(\frac{A_w^*}{m^*}, \omega^*)$ can be recovered by

$$\phi_1 = \frac{3}{2} \frac{s_0^*}{L_w^*}$$

The very high amplitudes ($\phi_1 \approx 60^\circ$) associated with low values of both $\frac{A_w^*}{m^*}$ and ω^* suggest such combinations probably do not occur, and that the parameter space can be further restricted. The data source for ω^* does not report measurements necessary to obtain $\frac{A_w^*}{m^*}$, but we could maybe find other sources. We can estimate the typical hover wingbeat amplitude by assuming that the lowest ω^* occur at the highest $\frac{A_w^*}{m^*}$. For this set of parameters ($\omega^* = 8$ and $\frac{A_w^*}{m^*} = 0.25$) the amplitude is $s_0^* \approx 0.23$, which corresponds to $\phi_1 \approx 26^\circ$.

Nonzero Mean Wing Pitch for Inclined Stroke Plane Hover

C_ψ may deviate from its “optimum” value. At this value, the wingbeat is symmetric, meaning the net force vector is normal to the stroke. This means hovering at the optimal C_ψ requires a horizontal stroke plane. Dragonflies however are known to hover with an inclined stroke plane, which could be due to efficiency considerations that we have ignored (picking the maximum C_ψ may not be exactly optimal energy-wise, even if it minimizes s_0 , or our model simply does not capture other real effects that lead to the inclined stroke plane being advantageous). Without going too deep into these energy considerations at least for now, it is interesting at this point to consider the effect of the pitch parameters not just on the force magnitude but on its

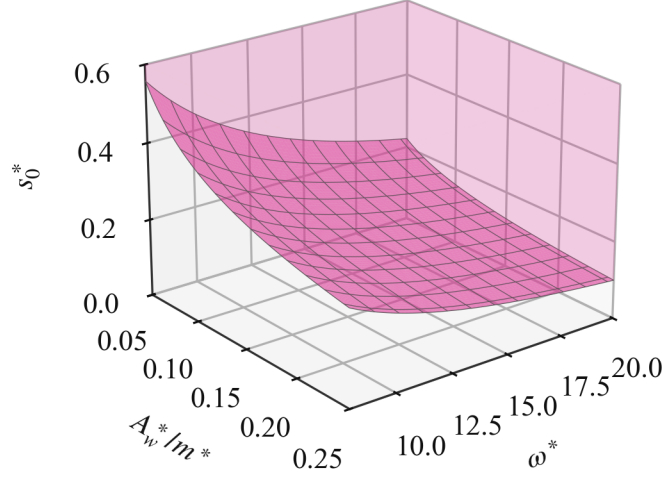


Figure 5: Approximate region of hover feasibility, with boundary surface.

direction relative to the stroke plane. With an inclined stroke (angle γ_0 to the horizontal), hover (or vertical flight) requires the net aerodynamic force vector to be angled by $-\gamma_0$ to the stroke plane normal.

We denote the force direction angle relative to the stroke plane with β . We find numerically (and could probably show analytically) that β depends only on the mean pitch angle ψ_0 , and on the sign of δ_0 . To be exact, there is a slight variation in $\beta(\psi_0)$ over the range of ψ_1 and $|\delta_0|$ on the order of 0.1° to 1° depending on the wingbeat amplitude. We also find that the (ψ_1, δ_0) that maximizes C_ψ is invariant with ψ_0 . We show $\beta(\psi_0)$ in Figure 6 along with the optimal C_ψ at a given ψ_0 .

Three observations from the plot: (a) β is monotonic in ψ_0 , so ψ_0 can be used conveniently as a control variable for the force direction, in addition to the stroke plane angle γ_0 , (b) a 90° turn in ψ_0 reverses the force direction, so only moderate changes in ψ_0 are needed to achieve relatively large changes in β (there is factor of ~ 2), and (c) the wing stroke is most efficient when the mean pitch attitude points out-of-plane or in-plane, and least efficient when it forms a 45° angle with the stroke plane.

Hover Control Variables

The above insights suggest a few possible ways to achieve hover control. Candidates for control variables can be split into two groups: those that control force magnitude (ϕ_1, ω^* through q^* and ψ_1, δ_0 through C_ψ), and those that control force direction (γ_0, ψ_0). Note this is not such a clean split, because ψ_0 also changes force magnitude through C_ψ , just more weakly than it changes direction. For simplicity, we would rather consider ω^* to be a fixed parameter, so we will exclude it at least for now. Since we also do not fully understand the factors that set the hovering stroke plane angle, we will also consider γ_0 to be fixed for now (later on, we compare the effectiveness of γ_0 and ψ_0 as directional control variables). In addition, it is beneficial to keep C_ψ at or near its maximum value, which sets ψ_1 and δ_0 . This leaves ϕ_1 (or s_0^*) for magnitude and ψ_0 for direction. At equilibrium, ψ_0 will be such that $\beta = -\gamma_0$ (force is vertical), and ϕ_1 such that $F_a^* = 1$. Table 2 shows the required value of ψ_0 for different γ_0 , for a reference configuration $L_w^* = 0.75$, $\frac{A_w^*}{m^*} = 0.15$, and

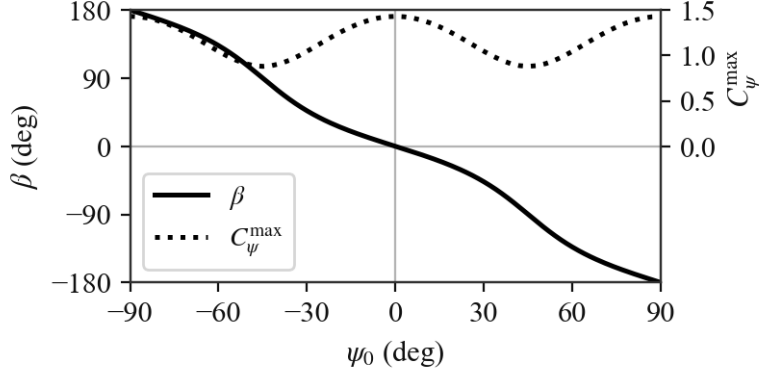


Figure 6: Effect of ψ_0 on force direction and wing stroke efficiency.

$\omega^* = 14$. Different values of these parameters impose a different ϕ_1 for hover, which results in small changes in ψ_0 up to a few percent.

Table 2: Mean pitch for hover at various stroke plane angles for reference configuration

γ_0	ψ_0	$C_\psi^{\max}$
0°	0°	1.43
10°	7.9°	1.40
20°	15.1°	1.31
30°	21.4°	1.21
40°	26.7°	1.11

ϕ_1 is set by

$$\phi_1^{\text{hover}} = \frac{3}{2} \left(L_w^* \omega^* \sqrt{C_\psi^{\max}(\psi_0) \frac{A_w^*}{m^*}} \right)^{-1}$$

The reference configuration used hereafter uses the following parameters (Table 3)

Table 3: Reference configuration

A_w^*/m^*	L_w^*	ω^*	γ_0
0.15	0.75	14	40°

2.2 Hover Dynamics and Control

Unsteady Dynamics

Before designing a controller, we look at how the actual, unsteady hover dynamics differ from the equilibrium picture we have been working with so far. Indeed the body velocity will not be exactly zero throughout, and will oscillate according to the fluctuations of the aerodynamic force over a wingbeat cycle. There is an additional effect due to small but nonzero mass of the wings, which will make the body bob up and down in

the direction of the flapping motion. We have neglected this in our work so far. In this part we will evaluate the relative importance of the two effects on the body oscillations.

A key parameter here is the fore/hind wing phase shift σ_0 . We expect the oscillations to be more pronounced when the wings beat in-phase ($\sigma_0 = 0^\circ$) than when they beat out-of-phase ($\sigma_0 = 180^\circ$). In fact, if the forewing and hindwing motions are identical except for the phase shift, the wing inertia effect will be zero (true for translation, not for body pitch, which we do not consider in the present model).

We introduce a relative wing mass parameter $m_w^* = m_w/m$. According to [1], values range from 0.01 to 0.03 for *Anisoptera*. We run one hover wingbeat to obtain the body oscillation peak-to-peak amplitude, comparing the ideal massless wing case to the $m_w^* = 0.02$ case, for the reference configuration of Table 3 (Figure 7). The wing mass and aerodynamic contributions to the body oscillations appear to be of the same order, except, as expected, when the wings beat out-of-phase. We will consider $m_w^* = 0$ in the following, but should keep in mind that this is only effectively true at $\sigma_0 = 180^\circ$.

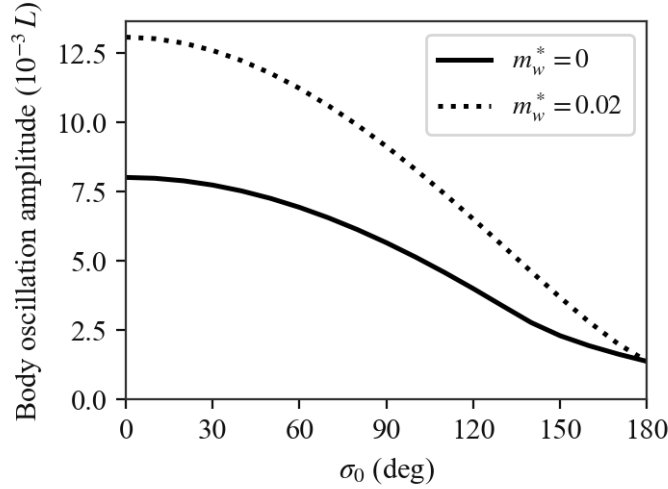


Figure 7: Effect of σ_0 and m_w^* on body oscillations during hover.

Another effect of the aerodynamic force oscillations is positional drift. For a constant, open-loop wingbeat pattern, the dragonfly settles in a steady-state that has a small but nonzero mean body velocity, when the equilibrium prediction is that the zero mean force condition occurs at exactly zero mean body velocity.

Overall these effects are relatively mild, with instantaneous body velocities of approximately $0.25s_0^*\omega^*$ in the worst case scenario ($\sigma_0 = 0^\circ$, $\omega^* = 8$, $A_w^*/m^* = 0.25$), and only about $\frac{1}{10}$ of that for $\sigma_0 = 180^\circ$, so the conclusions drawn from the equilibrium analysis should be applicable to controller design.

Hover Controller

Recall we identified ϕ_1 and ψ_0 as the two control variables for hover. ϕ_1 controls force magnitude, and ψ_0 controls force direction, with the secondary effect of reducing the force magnitude as it moves away from the stroke plane normal mean $\psi_0 = 0$. It seems natural to use ϕ_1 to control altitude, and ψ_0 to control fore-aft drift by tilting the force direction vector from the vertical. Specifically we propose the following control scheme

$$\begin{aligned}\psi_0 &= \psi_0^{\text{eq}} - K_x(x - x_0) \\ \phi_1 &= \phi_1^{\text{eq}} - K_z(z - z_0)\end{aligned}$$

Note that orienting the force in the $-x$ direction requires pitching up (increasing ψ_0), so we will need $K_x < 0$. To design the proportional gains for the reference configuration (Table 3) we begin by forming the Jacobian

matrix of the wingbeat-averaged aerodynamic force $\vec{F}_a^*$ at the hover equilibrium, using finite differences, to evaluate the control effectiveness of the two variables about the force axes. The Jacobian is

$$J = \begin{bmatrix} \frac{\partial F_{a,x}^*}{\partial \psi_0} & \frac{\partial F_{a,x}^*}{\partial \phi_1} \\ \frac{\partial F_{a,z}^*}{\partial \psi_0} & \frac{\partial F_{a,z}^*}{\partial \phi_1} \end{bmatrix} \approx \begin{bmatrix} -2.06 & 0.04 \\ -0.96 & 5.69 \end{bmatrix}$$

Notably ϕ_1 has a negligible effect on force tilt from the vertical ($F_{a,x}^*$), while as we pointed out before ψ_0 does have a significant secondary effect on $F_{a,z}^*$. Since ϕ_1 cleanly drives $F_{a,z}^*$, its control loop will absorb the ψ_0 cross term. The decoupled dynamics are (note mass matrix is I because of nondimensionalization)

$$\begin{aligned} \ddot{x}^* &= -2.06 \times \delta\psi_0 \\ \ddot{z}^* &= 5.69 \times \delta\phi_1 \end{aligned}$$

So $\omega_n^2 = -2.06K_x = 5.69K_z$. We want ω_n slow enough that the controller sees the wingbeat-averaged force and not the fluctuations. Using $\omega_n = 1.5$ (10 times smaller than the reference wingbeat frequency $\omega^* = 14$) we get

$$\begin{aligned} K_x &= -1.09 \\ K_z &= 0.39 \end{aligned}$$

The step response is shown in [Figure 8](#). The figure shows the reference configuration along with low and high wing loading configuration. A_w^*/m^* was chosen as the variable here because it had the greatest effect on the damping of the response. The response is very noticeably more damped as A_w^*/m^* increases. Over the range of parameters, using the same gains designed for the reference configuration, the achieved ω_n was close to the design value of 1.5, and the response did not vary very much except along the A_w^*/m^* axis. The controller was stable across the full parameter space.

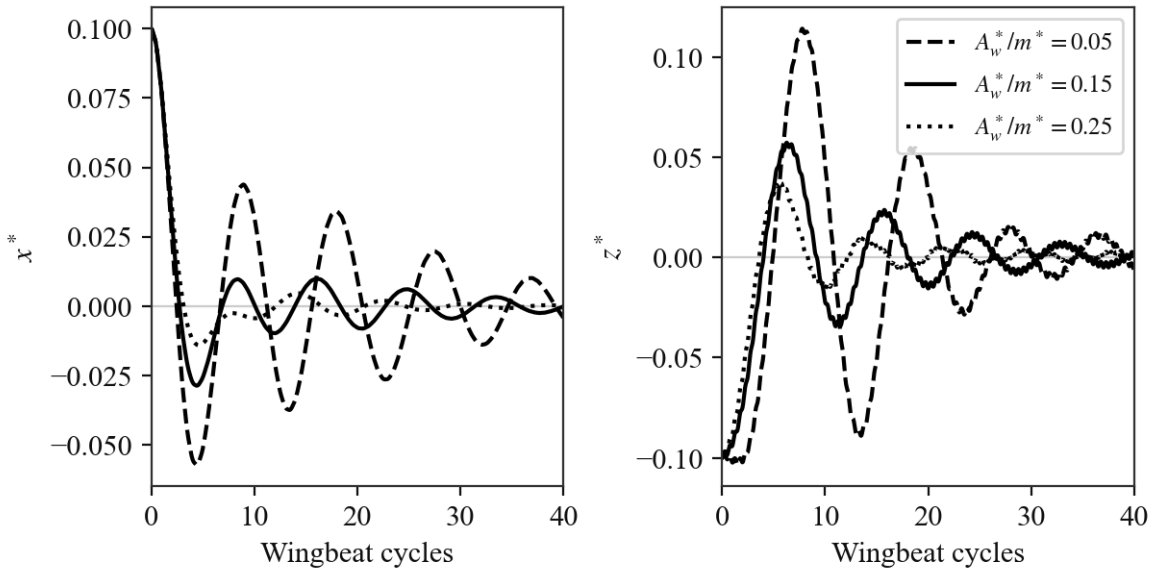


Figure 8: Effect of A_w^*/m^* on hover controller step response.

A more interesting result is the influence of stroke plane angle on damping, as shown in [Figure 9](#). It appears that damping is strongest within the stroke plane, which means that vertical damping in the $\gamma_0 = 0^\circ$

case is relatively low, and increases as the stroke plane is tilted. Horizontal damping decreases accordingly, with the end result being that there is a better match in the damping characteristics along both axes. Perhaps this is another advantage of hovering with an inclined stroke plane.

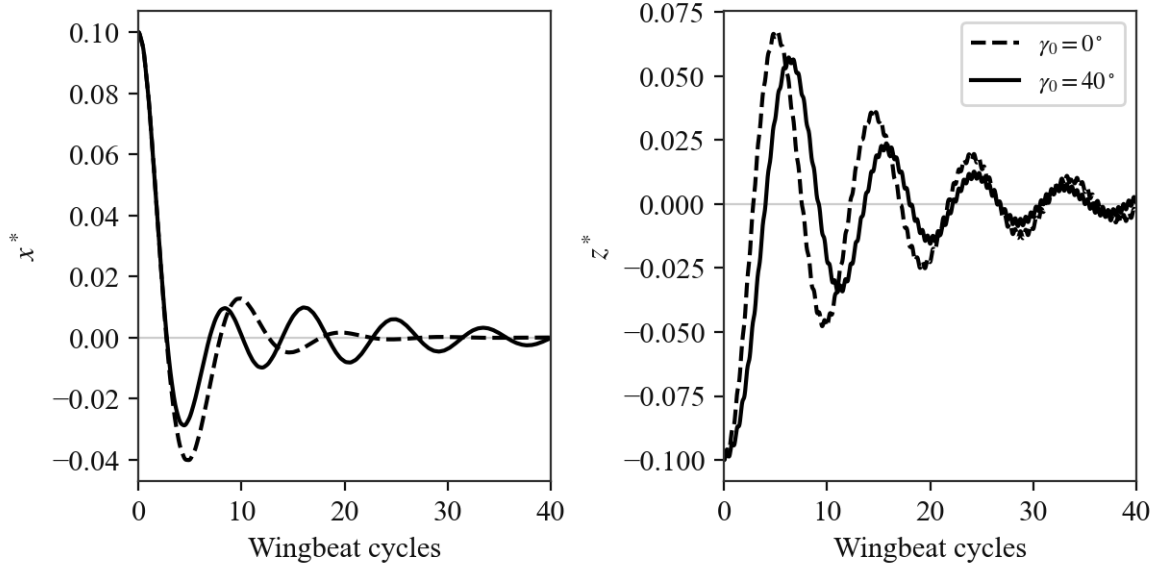


Figure 9: Effect of γ_0 on hover controller step response.

Robustness to Sensing Delay

We have assumed instant, perfect knowledge of body position. We now introduce a sensing delay and evaluate the constraints it imposes on controller design [needs to be written out].

3 Moving and Maneuvering Flight

So far we have focused on stationary flight, with at most small amplitude motions about a stationary or slowly drifting position. That is the case where the body velocity relative to the air is much less than the wing velocity relative to the body ($v_b^* \ll s_0^* \omega^*$). Since we are interested not just in hover but in maneuvering flight, we need to consider the case where $v_b^* \sim s_0^* \omega^*$, perhaps even $v_b^* \gg s_0^*$.

3.1 Equilibrium Solutions

[needs to be written out]

3.2 Control for Maneuvering Flight

[needs to be written out]

References

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